

ISE 5970: Advanced Systems Optimization

University of Oklahoma School of Industrial and Systems Engineering

Spring 2025

Instructions for Assignment Submission:

1. The assignment must be uploaded to Canvas by the indicated due date. The solution should be uploaded to Canvas as a PDF file.
2. Along with your main submission (in PDF format), please ensure you also attach any supplementary files (Excel, Python) that were instrumental in your work. If any Python code was used, upload it in HTML format.

Assignment 4: Lagrangian Relaxation and Strong Inequalities

1. **(30 points) Capacitated facility location:** A logistics company is considering opening warehouses at a set of candidate locations to serve customer demand in different zones. Each facility has a fixed opening cost and a limited capacity, and each customer zone has a known demand. The company must decide which facilities to open and how to assign customers to open facilities to minimize the total cost. Each customer must be fully served, and no facility can exceed its capacity.

Facility data:

Facility i	Fixed Cost f_i	Capacity s_i
1	100	20
2	80	15
3	90	25

Customer data:

Customer j	Demand d_j
A	10
B	10
C	15
D	10

Shipping costs c_{ij} :

Facility i	A	B	C	D
1	4	6	8	5
2	5	4	6	7
3	6	3	5	4

- (a) **(5 points) Concepts:** Explain the concept of a relaxation in the context of integer programming. What does it mean to relax a problem, and what role does this serve in optimization? Then, describe why a Lagrangian relaxation constitutes a valid relaxation of the original problem.
- (b) **(10 points) Model Formulation:** Formulate the Capacitated Facility Location Problem (CFLP) as a mixed-integer linear program. Clearly define all sets, parameters, and decision variables. Present the objective function and all relevant constraints. Use Gurobi to solve the model and report the optimal solution, including which facilities are opened, how customers are assigned, and the total cost.

- (c) **(10 points) Lagrangian Relaxation:** Choose a set of constraints to relax using a Lagrangian relaxation approach, and justify your choice. Derive the corresponding Lagrangian function and describe the resulting decomposition structure. Solve the relaxed problem for several iterations using sub-gradient optimization. For each iteration, report:
- the current values of the Lagrange multipliers,
 - the subproblem solutions,
 - the corresponding Lagrangian bound, and
 - any feasible solutions encountered.
- (d) **(5 points) Alternative Relaxations:** Discuss at least one alternative relaxation that could be applied to this problem. Clearly describe how the alternative approach differs from the one you implemented in part (b), and explain any potential advantages or disadvantages it might offer in terms of decomposition or solution quality.
2. **(40 points) Multicommodity Flow with Nonhomogeneous Goods:** A humanitarian organization is coordinating the delivery of three critical supplies—**food (F)**, **water (W)**, and **medical kits (M)**—from two supply hubs (nodes 1 and 2) to a field hospital located at node 7. The supplies must be routed through a constrained road network. Node 1 has a capacity to send up to 8 units of each commodity, while node 2 can send up to 7 units of each. The hospital at node 7 requires exactly 15 units of each type of supply.
- Nodes: $N = \{1, 2, 3, 4, 5, 6, 7\}$
 - Arcs: $A = \{(1, 3), (1, 4), (2, 3), (2, 5), (3, 4), (3, 5), (4, 6), (5, 6), (6, 7)\}$
 - Commodities: $K = \{F, W, M\}$ (Food, Water, Medical)

Arc Capacities and Shipping Costs:

Arc (i,j)	Capacity u_{ij}	Cost c_{ij}
(1,3)	60	3
(1,4)	45	2
(2,3)	30	4
(2,5)	30	1
(3,4)	45	2
(3,5)	30	3
(4,6)	30	2
(5,6)	30	2
(6,7)	60	1

- (a) **(10 points) Model Formulation:** Formulate the multicommodity minimum-cost flow problem using arc-based decision variables. Clearly define all sets, parameters, and decision variables. Write the complete objective function and constraints, including flow conservation and arc capacity limits. Find the optimum solution and report your findings.
- (b) **(15 points) Lagrangian Relaxation:** Apply any Lagrangian relaxation strategy of your choosing and argue your decision. Show the resulting decomposable structure, and explain how the subproblems can be interpreted. Use sub-gradient optimization to update the Lagrange multipliers and solve the problem to optimality. You can use Gurobi to solve the problem, but you must explain each iteration, including objective values, subproblem solutions, and sub-gradient updates.
- (c) **(15 points) Column Generation:** Propose a column generation strategy to solve the problem. *Suggestion:* You may use a path-based formulation. Define the restricted master problem (RMP) and the pricing subproblem clearly. Solve the problem to optimality using Gurobi, but explain each iteration of the algorithm, including how new columns are generated and how optimality is achieved.
- (d) **(10 points Extra Credit) Nonhomogeneous Arc Usage:** Suppose now that each commodity consumes a different amount of capacity per arc, as specified below. Use either the Lagrangian relaxation or column generation approach to solve this modified version of the problem. Describe clearly any adjustments made to the formulation, subproblems, or decomposition strategy.

Per-unit capacity consumption p_{ij}^k :

Arc	p_{ij}^F	p_{ij}^W	p_{ij}^M
(1,3)	1	2	3
(1,4)	1	1	2
(2,3)	2	1	2
(2,5)	1	2	1
(3,4)	1	1	1
(3,5)	2	1	3
(4,6)	1	2	1
(5,6)	1	1	2
(6,7)	1	1	1

3. **(30 points) Generalized Assignment Problem (GAP):** There are $m = 10$ workers and $n = 5$ products. Each worker $i \in \{1, \dots, m\}$ must be assigned to produce exactly one product. Due to differences in skill levels, the amount of resource required by worker i to produce product j is a_{ij} . Each product j has a total available resource limited by b_j , and it can be produced by multiple workers, as long as the total resource consumption by the assigned workers does not exceed b_j .

The assignment of worker i to product j yields a profit c_{ij} , and the objective is to maximize the total profit across all assignments. The decision variable x_{ij} is binary and equals 1 if worker i is assigned to produce product j , and 0 otherwise.

The GAP formulation is:

$$\begin{aligned}
& \text{maximize} && \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\
& \text{subject to} && \sum_{j=1}^n x_{ij} = 1, \quad \forall i = 1, \dots, m \\
& && \sum_{i=1}^m a_{ij} x_{ij} \leq b_j, \quad \forall j = 1, \dots, n \\
& && x_{ij} \in \{0, 1\}, \quad \forall i, j
\end{aligned}$$

where $x_{ij} = 1$ if agent i is assigned to task j , and 0 otherwise.

The available resource vector is:

$$b = (80, 63, 75, 98, 59)$$

The resource consumption matrix $A = (a_{ij})$ and profit matrix $C = (c_{ij})$ are:

$$A = \begin{pmatrix} 3 & 24 & 53 & 27 & 17 \\ 15 & 23 & 43 & 74 & 23 \\ 54 & 43 & 27 & 21 & 36 \\ 92 & 83 & 45 & 35 & 26 \\ 19 & 10 & 33 & 43 & 12 \\ 91 & 55 & 32 & 26 & 23 \\ 15 & 25 & 36 & 37 & 28 \\ 47 & 43 & 33 & 28 & 23 \\ 34 & 25 & 32 & 46 & 43 \\ 35 & 23 & 34 & 25 & 40 \end{pmatrix} \quad C = \begin{pmatrix} 15 & 44 & 76 & 43 & 34 \\ 19 & 23 & 45 & 46 & 34 \\ 10 & 6 & 3 & 23 & 15 \\ 60 & 45 & 34 & 36 & 23 \\ 11 & 12 & 34 & 44 & 10 \\ 67 & 65 & 34 & 20 & 37 \\ 23 & 34 & 24 & 47 & 27 \\ 23 & 25 & 35 & 15 & 27 \\ 12 & 13 & 24 & 25 & 24 \\ 10 & 15 & 23 & 12 & 13 \end{pmatrix}$$

- (10 points) Formulate and solve the problem as a mixed integer programming (MIP) model using branch-and-bound. How many nodes were required to explore before finding the optimal solution?
- (10 points) Identify and add at least two strong valid inequalities (e.g., cover inequalities) that help strengthen the LP relaxation.
- (10 points) Compare the number of branch-and-bound nodes explored with and without these valid inequalities. In your words, explain the impact of the added inequalities on the solver's performance.